

CHAPTER 6: GEOMETRICAL OPTICS

At the end of this chapter, you'll be able to:

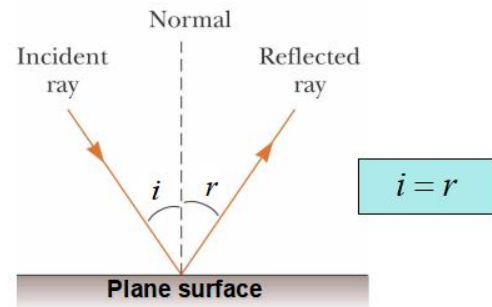
SUBTOPIC	LEARNING OUTCOMES
6.1	Reflection at a spherical surface a) State and use radius of curvature, $R = 2f$ for spherical mirror. b) Sketch ray diagrams with a minimum of two rays to state the characteristics of image formed by spherical mirrors. c) Use mirror equation, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ for real object only. <i>*sign convention for focal length, f and radius of curvature, R: positive f and R for concave mirror; and negative f and R for convex mirror.</i> d) Use magnification, $m = \frac{h_i}{h_o} = -\frac{v}{u}$ respectively.
6.2	Thin lenses a) Sketch ray diagrams with a minimum of two rays to state the characteristics of image formed by concave and convex lenses. b) Use thin lens equation, $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ for real object only. <i>*sign convention for focal length, f: positive f for convex lens; and negative f for concave lens.</i> c) Use magnification, $m = \frac{h_i}{h_o} = -\frac{v}{u}$ respectively.

6.1 REFLECTION AT A SPHERICAL SURFACE

Laws of Reflection

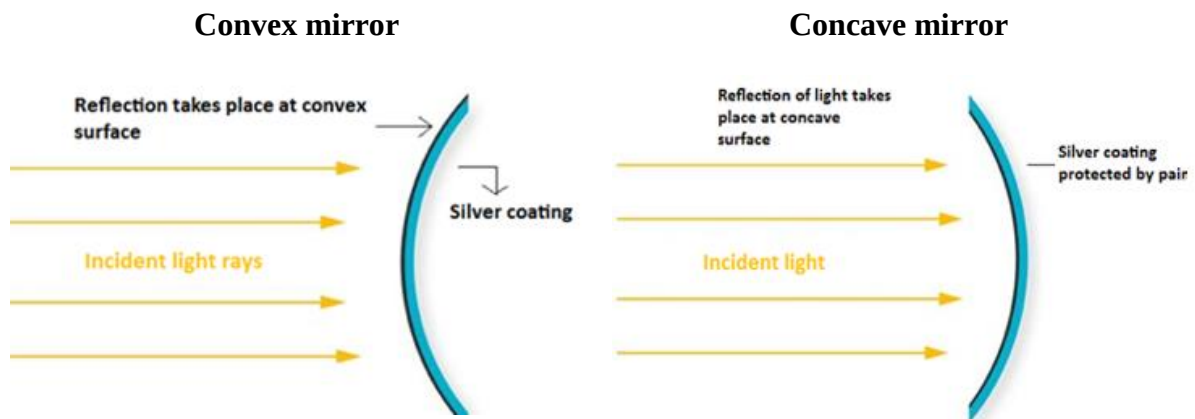
There are two laws of reflection of light:

- The angle of reflection θ_r , equals the angle of incidence θ_i
- The incident ray, the normal (perpendicular to surface), and the reflected ray all lie in the same plane.

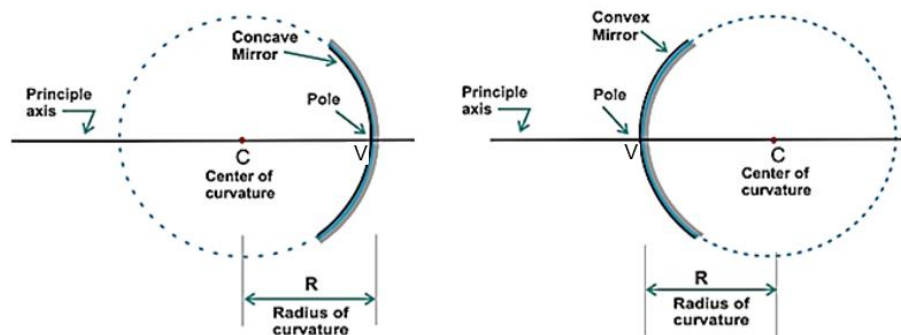


Spherical Mirror

- Spherical mirror is defined as a reflecting surface that is part of a sphere.
- There are two types of spherical mirror:
 - Convex** (curving outwards) mirror.
 - Concave** (curving inwards) mirror.

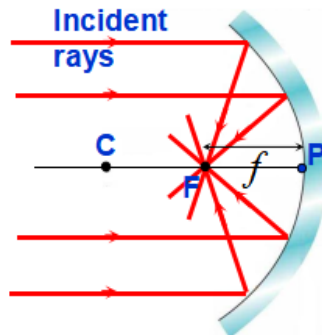


- Some terms of spherical mirror:
 - Centre of curvature (point C) is defined as the center of the sphere of which a curved mirror forms a part.
 - Radius of curvature (point R) is defined as distance from the pole, V to the center of curvature, C.
 - Pole or vertex (point V) is defined as point at the center of the mirror.
 - Principal axis is defined as straight line passing through V and C of a spherical mirror.



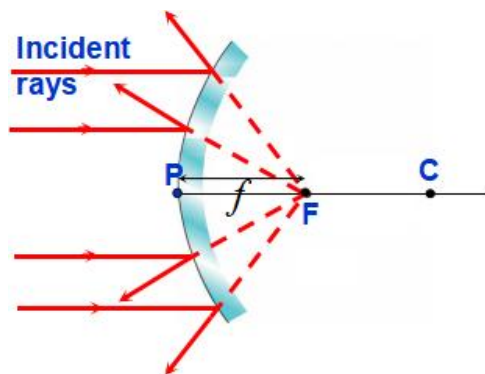
Focal Point and Focal Length

1. **Focal Point or Focus, F** is defined as a point at midway between center of curvature and pole.



(a) Concave mirror / Converging mirror

2. For concave mirror, focal point is a point where the incident parallel rays converge after reflection of the mirror. Its focal point is real.

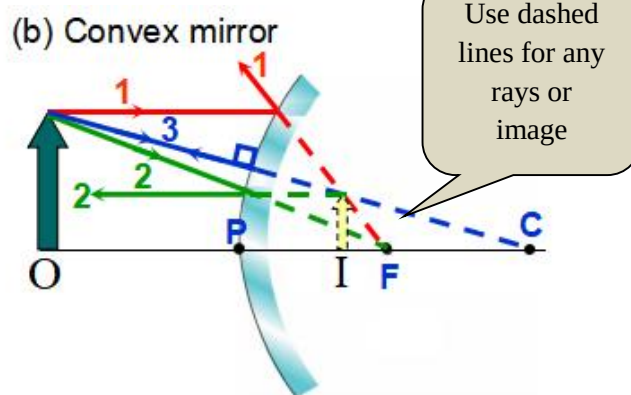
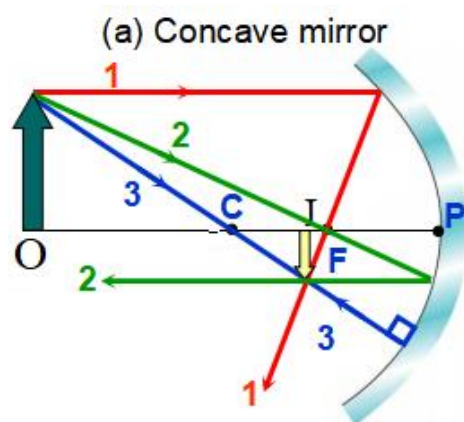


(b) Convex mirror / Diverging mirror

3. For convex mirror, focal point is a point where the incident parallel rays seem to diverge from a point behind the mirror after reflection. Its focal point is virtual.
4. **Focal length, f** is defined as the distance between the focal point (focus) and the pole P of the spherical mirror.
5. **Radius of curvature** for a spherical mirror, $R = 2f$

Ray Diagrams for Spherical Mirror

Ray diagram is the simple graphical method to indicate the positions of the object in a system of mirrors or lenses. Ray diagrams below showing the graphical method of locating an image formed by concave and convex mirror.



Ray 1 - Parallel to principal axis, after reflection, passes through the focal point (focus) F of a concave mirror or appears to come from the focal point F of a convex mirror.

Ray 2 - Passes or directed towards focal point F reflected parallel to principal axis.

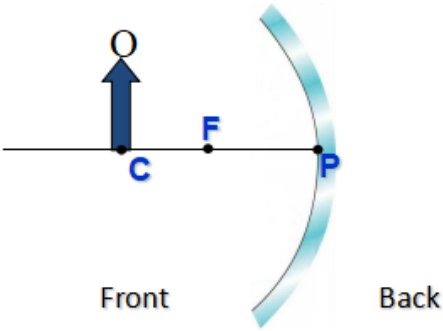
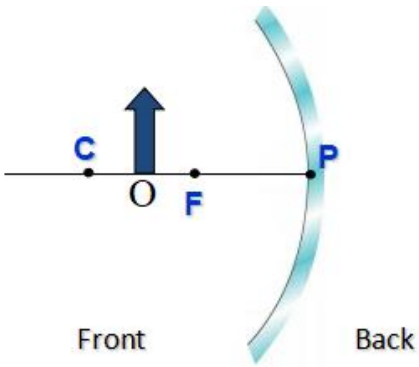
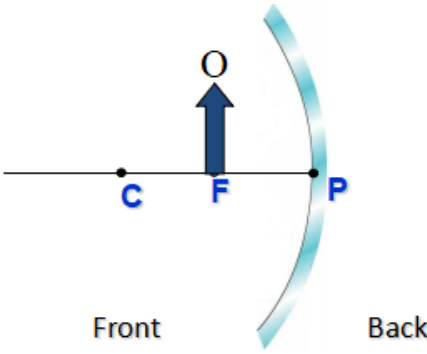
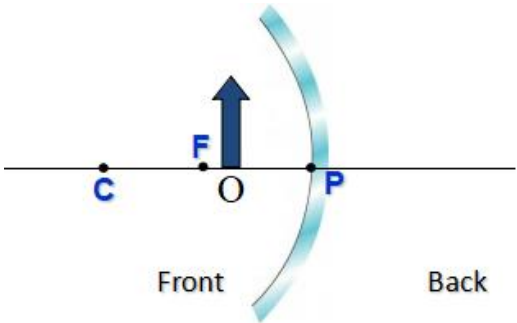
Ray 3 - Passes or directed towards centre of curvature C, reflected back along the same path.

Images formed by convex mirror

Object Distance, u	Ray Diagram	Image Characteristics
any position in front of the mirror		● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?

Images formed by concave mirror

Object Distance, u	Ray Diagram	Image Characteristics
$u > r$		● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?

$u = r$	 Front Back	● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?
$f < u < r$	 Front Back	● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?
$u = f$	 Front Back	● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?
$u < f$	 Front Back	● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?

Linear Magnification of the spherical mirror, M

$$M = \frac{h_i}{h_o} = -\frac{v}{u}$$

h_i : Image height
 h_o : Object height
 v : Image distance
 u : Object distance

$M > 1$	Magnified
$M = 1$	Same size
$M < 1$	Diminished

Equation of Spherical Mirror

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

f : Focal length
 u : Object distance
 v : Image distance

Sign convention for equation of spherical mirror

Physical Quantity	Positive (+)	Negative (-)
Object distance, u	Real Object (in front of the mirror)	Virtual object (at the back of the mirror)
Image distance, v	Real Image (in front of the mirror)	Virtual Image (at the back of the mirror)
Focal length, f	Concave mirror (converging)	Convex mirror (diverging)
Linear magnification, M	Upright	Inverted

Note:

In problem solving, the appropriate sign convention for the known quantities must be put into equation.

The final answer will also have a sign, which gives additional information about the image.

Real image is formed by the actual light rays that pass through the image.

Virtual image is formed when the light rays do not pass through the image point but appear to diverge from that point.

Example 1

An object is placed 10.0 cm in front of a concave mirror whose focal length is 15.0 cm.

Determine

- The position of the image.
- The linear magnification and state the properties of the image.

Answers:

$$u = +10.0 \text{ cm}; f = +15.0 \text{ cm}$$

a)

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{+15} = \frac{1}{+10} + \frac{1}{v}$$

$$v = -30.0 \text{ cm}$$

b)

$$m = -\frac{v}{u} = -\frac{(-30.0)}{+10.0}$$

$$m = 3$$

virtual, upright, magnified

Example 2

A woman who is 1.5 m tall is located 3.0 m from a convex antishoplifting mirror. The focal length of the mirror is 0.25 m. Find:

- The radius of curvature of the mirror.
- The position of her image.
- The magnification.
- The height of the image.
- State the characteristics of the image formed.

Answers:

$$h_o = 1.5 \text{ m}; u = +3.0 \text{ m}; f = -0.25 \text{ m}$$

a)

$$R = 2f$$

$$R = 2(-0.25)$$

$$R = -0.5 \text{ m}$$

b)

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{(-0.25)} = \frac{1}{+3.0} + \frac{1}{v}$$

$$v = -0.231 \text{ m}$$

c)

$$m = -\frac{v}{u}$$

$$m = -\frac{(-0.231)}{+3.0}$$

$$m = 0.077$$

d)

$$m = \frac{h_i}{h_o}$$

$$0.077 = \frac{h_i}{1.5} \rightarrow h_i = 0.12 \text{ m}$$

e)

Virtual, upright, diminished.

6.2 THIN LENSES

1. Thin lens is a transparent material with two spherical refracting surfaces whose thickness is thin compared to the radii of curvature of the two refracting surfaces.
2. There are two types of thin lenses, convex (converging) lens and concave (diverging) lens. Figure shows various types of thin lenses.

Converging (Convex) lenses



Biconvex



Plano-convex



Convex meniscus

Diverging (Concave) lenses



Biconcave



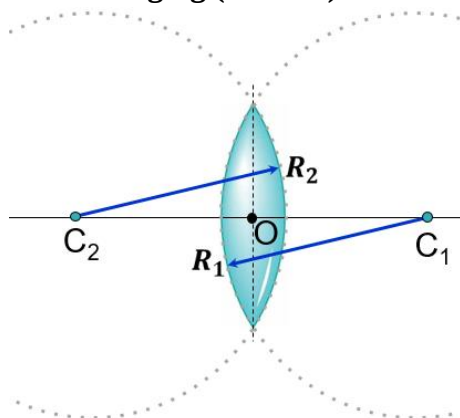
Plano-concave



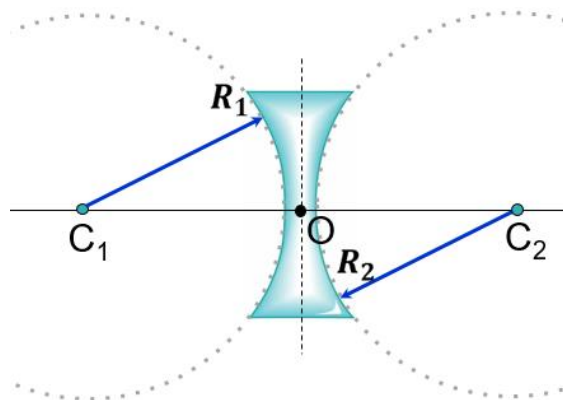
Concave meniscus

3. Some terms of spherical mirror:

Converging (Convex) lenses



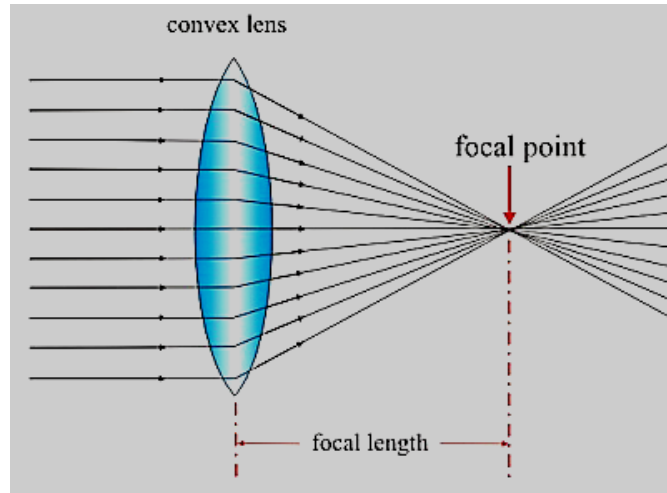
Diverging (Concave) lenses



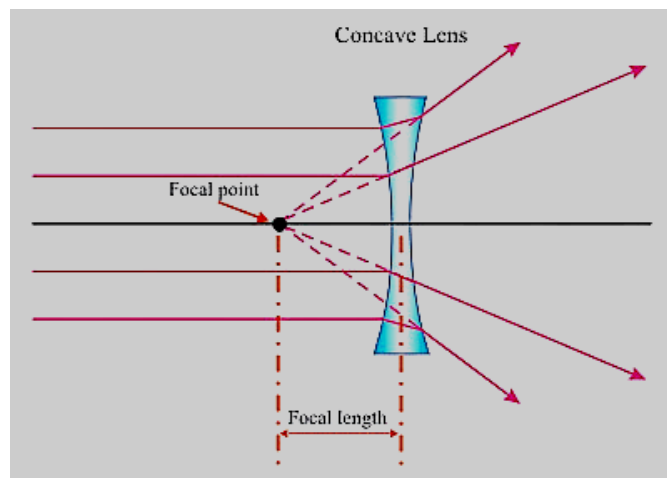
- i. Centre of Curvature (C_1 and C_2) is defined as the centre of sphere of which the surface of the lens is a part.
- ii. Radius of Curvature (R_1 and R_2) is defined as the radius of the sphere of which the surface of the lens is a part.
- iii. Principal axis, P is defined as a line through the center of curvature C_1 & C_2 .
- iv. Optical Centre (Point O) is defined as a point where all rays pass through this point continue in a straight line.

Focal Point and Focal Length

1. **Focal point, F** is defined as a point where the parallel rays incident on the lens either converge after refraction for a convex lens or diverge after refraction for a concave lens.



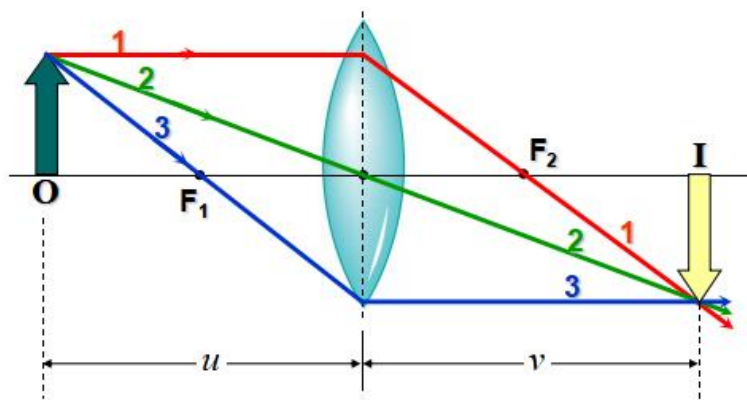
2. For convex (converging) lens, **focus point** is the point on the principal axis where rays which are parallel and close to the principal axis converge after passing through the lens.



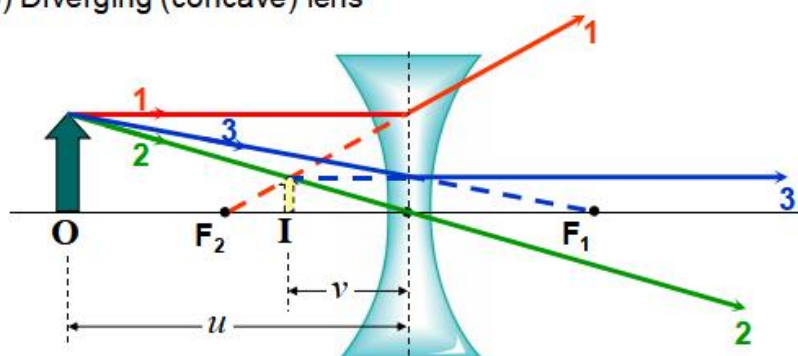
3. For concave (diverging) lens, **focus point** for concave (diverging) lens is the point on the principal axis where rays which are parallel to the principal axis seem to diverge from after passing through the lens.
4. **Focal length** is the distance between focus point, F to the optical centre of the lens, O.

Ray Diagrams for Lenses

(a) Converging (convex) lens



(b) Diverging (concave) lens



Ray 1 - Parallel to the principal axis, after refraction by the lens, passes through the focal point (focus) F_2 of a converging lens or appears to come from the focal point F_2 of a diverging lens.

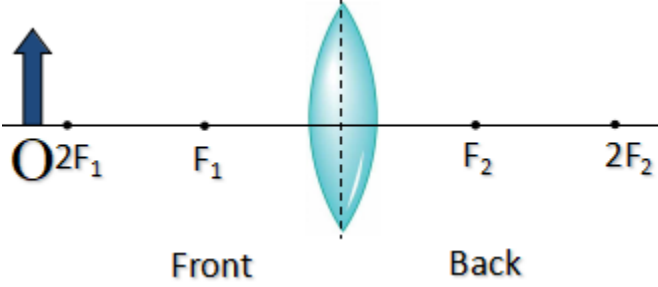
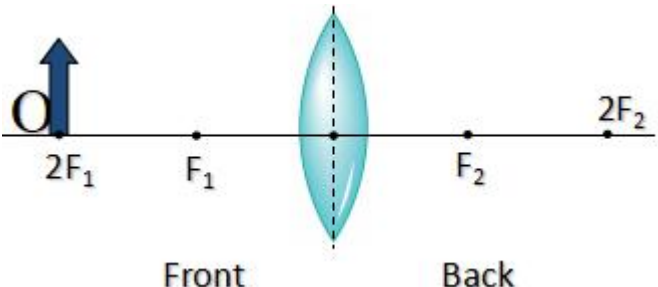
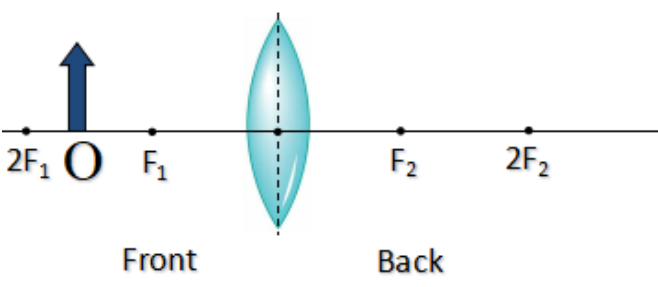
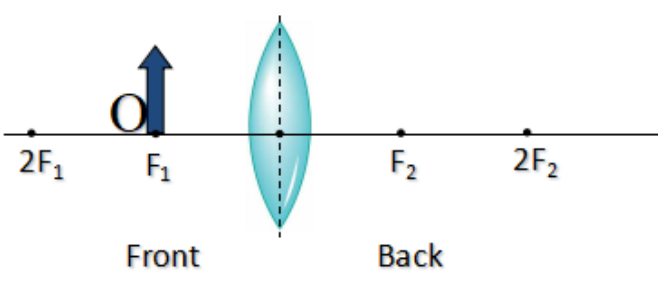
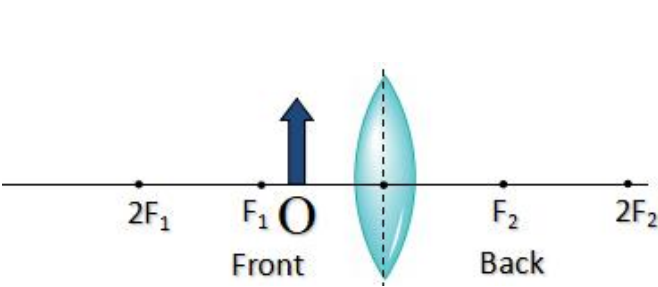
Ray 2 - Passes through the optical centre of the lens is undeviated.

Ray 3 - Passes through the focus F_1 of a converging lens or appears to converge towards the focus F_1 of a diverging lens, after refraction by the lens the ray parallel to the principal axis.

Image formed by Concave (Diverging) Lens

Object Distance, u	Ray Diagram	Image Characteristics
any position in front of the mirror		● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?

Image formed by Convex (Converging) Lens

Object Distance, u	Ray Diagram	Image Characteristics
$u > 2f$		● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?
$u = 2f$		● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?
$f < u < 2f$		● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?
$u = f$		● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?
$u < f$		● Real or Virtual? ● Upright or Inverted? ● Magnified or Diminished or Same Size? ● Position of image formed?

Linear Magnification of the spherical mirror, M

$$M = \frac{h_i}{h_o} = -\frac{v}{u}$$

 h_i : Image height h_o : Object height v : Image distance u : Object distance

$M > 1$	Magnified
$M = 1$	Same size
$M < 1$	Diminished

Thin lens equation

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

 f : Focal length u : Object distance v : Image distance**Sign convention for equation of thin lens**

Physical Quantity	Positive (+)	Negative (-)
Object distance, u	Real Object (in front of the lens)	Virtual object (at the back of the lens)
Image distance, v	Real Image (at the back of the lens)	Virtual Image (in front of the lens)
Focal length, f	Converging lens	Diverging lens
Radius of curvature, r	Centre of curvature is located in denser medium (convex surface)	Centre of curvature is located in less dense medium

Note:

In problem solving, the appropriate sign convention for the known quantities must be put into equation.

The final answer will also have a sign, which gives additional information about the image.

Example 3

An object is placed 30.0 cm to the left of a diverging lens of focal length 20.0 cm. Determine The location of image formed and the characteristics of the image.

$$f = -20.0 \text{ cm}; u = +30.0 \text{ cm}$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{-20.0} = \frac{1}{+30.0} + \frac{1}{v}$$

$$v = -12.0 \text{ cm} \text{ (image always upright, virtual \& diminished)}$$

Example 4

A convex lens with a focal length of 90.0 cm forms an image of a 3.20 cm tall real object that is to the left of the lens. The image is 4.50 cm tall and inverted. Find

- The object position from the lens.
- The image position from the lens. Is the image real or virtual?

Answers:

$$f = +90.0 \text{ cm}; h_o = 3.20 \text{ cm}; h_i = 4.50 \text{ cm}$$

a)

$$\frac{h_i}{h_o} = \frac{v}{u}$$

$$\frac{4.50}{3.20} = \frac{v}{u}$$

$$v = 1.406u$$

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$\frac{1}{+90.0} = \frac{1}{u} + \frac{1}{1.406u}$$

$$u = +154.0 \text{ cm}$$

b)

$$v = 1.406u$$

$$v = 1.406(154.0)$$

$$v = +216.52 \text{ cm}$$

Real image

Chapter 6 MCQs Practice:**SELECTED FORMULAE**

$$1. R = 2f$$

$$2. \frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

$$3. m = \frac{h_i}{h_o} = -\frac{v}{u}$$

DIY Quiz

Instruction:

Answer all questions in 10 minutes.

No discussion & do not refer to any notes.

You can only refer to the formulae given below.

- A concave mirror has a radius of curvature 30 cm. An object of height 12 cm forms an enlarged image 20 cm from the mirror.
 - Calculate the object distance.
 - Calculate the height of the image.
 - State the TWO characteristics of the image.

Answers:

$$a) u = 8.57 \text{ cm}$$

$$b) h_i = 28.0 \text{ cm}$$

c) DIY

PAST YEAR QUESTIONS

An object is placed at a distance of 4.0 cm from a concave mirror of focal length 10.0 cm.

- a) Calculate the image distance.
- b) Calculate the linear magnification.
- c) Determine TWO characteristics of the image formed.
- d) Sketch a labelled ray diagram to show the formation of the image.

[9 marks]